Exercise: add an element to the <u>front</u> of a vector.

E.g. if $V = [1, 2, 3]$, then after

```
pushfront(V, 7);
```

$V = [7, 1, 2, 3]$.

$V[i] = V[i-1]$

Prototype:

```
void pushfront(vector<int>& V, int x);
```

Sketch:

1. expand size of V by 1.
2. move all elements right by one space.
3. `V[0] = x;`

1. ~~`V.size()++;`~~ `V.resize(V.size()+1);`

   or, `V.push_back(0);`

2. `V[i] = V[i-1]`, from back → front

Note: repeated calls to push_front are quite costly!

Reason:

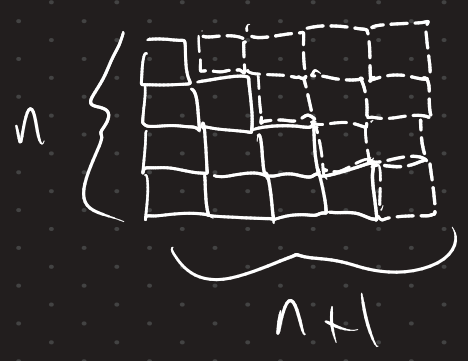
first time: 1 step
second time: 2 steps
third time: 3 steps
⋮

Total cost for n elements:

$$1+2+3+\cdots+n = \sum_{i=1}^{n} i$$

$$= \frac{n(n+1)}{2}$$

Aside: reason behind formula:



$$\Rightarrow \frac{n(n+1)}{2}$$

<u>Takeaway</u>: only efficient to add/remove elements from <u>back</u> of a vector!

Exercise: given a vector of integers & a target value t, write a function that decides if $\exists$ indexes i & j s.t.

"there exists" "such that"

$$V[i] + V[j] == t$$